Q1. [14 pts] True/False

(a) [1 pt] In Support Vector Machines, we maximize $\frac{\|w\|^2}{2}$ subject to the margin constraints.

○ True ● False

(b) [1 pt] In kernelized SVMs, the kernel matrix **K** has to be positive definite.

○ True ● False

(c) [1 pt] If two random variables are independent, then they have to be uncorrelated.

● True ○ False

(d) [1 pt] Isocontours of Gaussian distributions have axes whose lengths are proportional to the eigenvalues of the covariance matrix.

○ True ● False

(e) [1 pt] The RBF kernel ($K(x_i, x_j) = exp(-\gamma\|x_i - x_j\|^2)$) corresponds to an infinite dimensional mapping of the feature vectors.

● True ○ False

(f) [1 pt] If (X, Y) are jointly Gaussian, then X and Y are also Gaussian distributed.

● True ○ False

(g) [1 pt] A function $f(x, y, z)$ is convex if the Hessian of f is positive semi-definite.

● True ○ False

(h) [1 pt] In a least-squares linear regression problem, adding an L_2 regularization penalty cannot decrease the L_2 error of the solution w on the training data.

● True ○ False

(i) [1 pt] In linear SVMs, the optimal weight vector w is a linear combination of training data points.

● True ○ False

(j) [1 pt] In stochastic gradient descent, we take steps in the exact direction of the gradient vector.

○ True ● False

(k) [1 pt] In a two class problem when the class conditionals $P(x|y = 0)$ and $P(x|y = 1)$ are modelled as Gaussians with different covariance matrices, the posterior probabilities turn out to be logistic functions.

○ True ● False

(l) [1 pt] The perceptron training procedure is guaranteed to converge if the two classes are linearly separable.

● True ○ False

(m) [1 pt] The maximum likelihood estimate for the variance of a univariate Gaussian is unbiased.

○ True ● False

(n) [1 pt] In linear regression, using an L_1 regularization penalty term results in sparser solutions than using an L_2 regularization penalty term.

● True ○ False

Q2. [21 pts] Multiple Choice Questions

(a) [2 pts] If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = aX + b$, then the variance of Y is:

○ $a\sigma^2 + b$ ○ $a^2\sigma^2 + b$ ○ $a\sigma^2$ ● $a^2\sigma^2$

(b) [2 pts] In soft margin SVMs, the slack variables ξ_i defined in the constraints $y_i(w^T x_i + b) \geq 1 - \xi_i$ have to be

○ < 0 ○ ≤ 0 ○ > 0 ● ≥ 0

(c) [4 pts] Which of the following transformations when applied on $X \sim \mathcal{N}(\mu, \Sigma)$ transforms it into an axis aligned Gaussian? ($\Sigma = UDU^T$ is the spectral decomposition of Σ)

● $U^{-1}(X - \mu)$ ● $(UD)^{-1}(X - \mu)$ ○ $UD(X - \mu)$
● $(UD^{1/2})^{-1}(X - \mu)$ ○ $U(X - \mu)$ ○ $\Sigma^{-1}(X - \mu)$

(d) [2 pts] Consider the sigmoid function $f(x) = 1/(1 + e^{-x})$. The derivative $f'(x)$ is

○ $f(x) \ln f(x) + (1 - f(x)) \ln(1 - f(x))$ ● $f(x)(1 - f(x))$

○ $f(x) \ln(1 - f(x))$ ○ $f(x)(1 + f(x))$

(e) [2 pts] In regression, using an L_2 regularizer is equivalent to using a _____ prior.

○ Laplace, $2\beta\exp(-|x|/\beta)$ ○ Exponential, $\beta \exp(-x/\beta)$, for $x > 0$

● Gaussian with $\Sigma = cI, c \in R$ ○ Gaussian with diagonal covariance ($\Sigma \neq cI, c \in R$)

(f) [2 pts] Consider a two class classification problem with the loss matrix given as $\begin{pmatrix} \lambda_{11} & \lambda_{12} \\ \lambda_{21} & \lambda_{22} \end{pmatrix}$. Note that λ_{ij} is the loss for classifying an instance from class j as class i. At the decision boundary, the ratio $\frac{P(\omega_2|x)}{P(\omega_1|x)}$ is equal to:

○ $\frac{\lambda_{11} - \lambda_{22}}{\lambda_{21} - \lambda_{12}}$ ● $\frac{\lambda_{11} - \lambda_{21}}{\lambda_{22} - \lambda_{12}}$ ○ $\frac{\lambda_{11} + \lambda_{22}}{\lambda_{21} + \lambda_{12}}$ ○ $\frac{\lambda_{11} - \lambda_{12}}{\lambda_{22} - \lambda_{21}}$

(g) [2 pts] Consider the L_2 regularized loss function for linear regression $L(w) = \frac{1}{2}\|Y - Xw\|^2 + \lambda\|w\|^2$, where λ is the regularization parameter. The Hessian matrix $\nabla_w^2 L(w)$ is

○ $X^T X$ ○ $2\lambda X^T X$ ● $X^T X + 2\lambda I$ ○ $(X^T X)^{-1}$

(h) [2 pts] The geometric margin in a hard margin Support Vector Machine is

○ $\frac{\|w\|^2}{2}$ ○ $\frac{1}{\|w\|^2}$ ● $\frac{2}{\|w\|}$ ○ $\frac{2}{\|w\|^2}$

(i) [3 pts] Which of the following functions are convex?

○ $\sin(x)$ ● $|x|$ ○ $\min(f_1(x), f_2(x))$, where f_1 and f_2 are convex ● $\max(f_1(x), f_2(x))$, where f_1 and f_2 are convex

Q1. [10 pts] True or False

(a) [1 pt] The hyperparameters in the regularized logistic regression model are η (learning rate) and λ (regularization term).

○ True ● False

(b) [1 pt] The objective function used in L2 regularized logistic regression is convex.

● True ○ False

(c) [1 pt] In SVMs, the values of α_i for non-support vectors are 0.

● True ○ False

(d) [1 pt] As the number of data points approaches ∞, the error rate of a 1-NN classifier approaches 0.

○ True ● False

(e) [1 pt] Cross validation will guarantee that our model does not overfit.

○ True ● False

(f) [1 pt] As the number of dimensions increases, the percentage of the volume in the unit ball shell with thickness ϵ grows.

● True ○ False

(g) [1 pt] In logistic regression, the Hessian of the (non regularized) log likelihood is positive definite.

○ True ● False

(h) [1 pt] Given a binary classification scenario with Gaussian class conditionals and equal prior probabilities, the optimal decision boundary will be linear.

○ True ● False

(i) [1 pt] In the primal version of SVM, we are minimizing the Lagrangian with respect to w and in the dual version, we are minimizing the Lagrangian with respect to α.

○ True ● False

(j) [1 pt] For the dual version of soft margin SVM, the α_i's for support vectors satisfy $\alpha_i > C$.

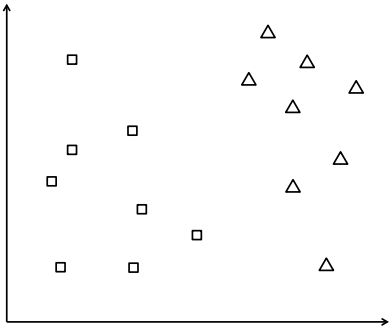
○ True ● False

Q2. [24 pts] Multiple Choice

(a) [3 pts] Consider the binary classification problem where $y \in \{0, 1\}$ is the label and we have prior probability $P(y = 0) = \pi_0$. If we model $P(x|y = 1)$ to be the following distributions, which one(s) will cause the posterior $P(y = 1|x)$ to have a logistic function form?

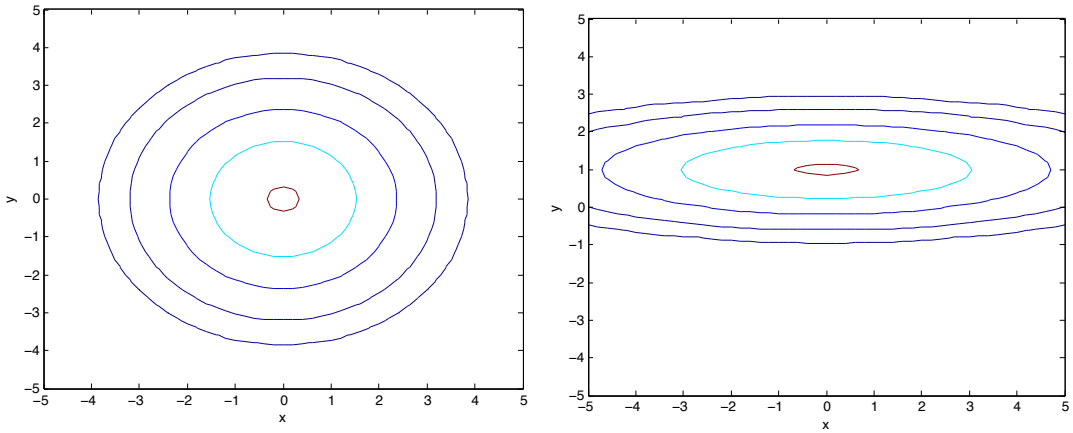
- ● Gaussian
- ○ Uniform
- ● Poisson
- ○ None of the above

(b) [3 pts] Given the following data samples (square and triangle belong to two different classes), which one(s) of the following algorithms can produce zero training error?



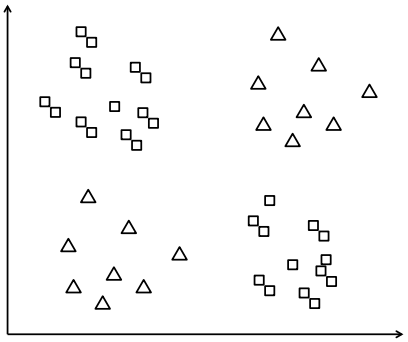
- ● 1-nearest neighbor
- ● Logistic regression
- ● Support vector machine
- ● Linear discriminant analysis

(c) [3 pts] The following diagrams show the iso-probability contours for two different 2D Gaussian distributions. On the left side, the data $\sim N(\mathbf{0}, \mathbf{I})$ where $\mathbf{I}$ is the identity matrix. The right side has the same set of contour levels as left side. What is the mean and covariance matrix for the right side's multivariate Gaussian distribution?



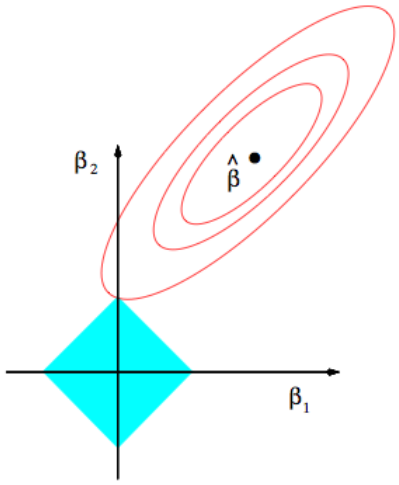
- ○ $\mu = [0, 0]^T$, $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- ● $\mu = [0, 1]^T$, $\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 0.25 \end{bmatrix}$
- ○ $\mu = [0, 1]^T$, $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- ○ $\mu = [0, 1]^T$, $\Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}$

(d) [3 pts] Given the following data samples (square and triangle mean two classes), which one(s) of the following kernels can we use in SVM to separate the two classes?



- ○ Linear kernel
- ● Gaussian RBF (radial basis function) kernel
- ● Polynomial kernel
- ○ None of the above

(e) [3 pts] Consider the following plots of the contours of the unregularized error function along with the constraint region. What regularization term is used in this case?



- ○ L_2
- ○ L_∞
- ● L_1
- ○ None of the above

(f) [3 pts] Suppose we have a covariance matrix

$$\Sigma = \begin{bmatrix} 5 & a \\ a & 4 \end{bmatrix}$$

What is the set of values that a can take on such that Σ is a valid covariance matrix?

- ○ $a \in \Re$
- ○ $a \geq 0$
- ● $-\sqrt{20} \leq a \leq \sqrt{20}$
- ○ $-\sqrt{20} < a < \sqrt{20}$

(g) [3 pts] The soft margin SVM formulation is as follows:

$$\min \frac{1}{2}\mathbf{w}^T\mathbf{w} + C\sum_{i=1}^{N}\xi_i$$
$$\text{subject to} \quad y_i(\mathbf{w}^T\mathbf{x_i} + b) \geq 1 - \xi_i \quad \forall i$$
$$\xi_i \geq 0 \quad \forall i$$

What is the behavior of the width of the margin ($\frac{2}{\|w\|}$) as $C \to 0$?

- ◯ Behaves like hard margin
- ◯ Goes to zero
- ● Goes to infinity
- ◯ None of the above

(h) [3 pts] In Homework 4, you fit a logistic regression model on spam and ham data for a Kaggle Competition. Assume you had a very good score on the public test set, but when the GSIs ran your model on a private test set, your score dropped a lot. This is likely because you overfitted by submitting multiple times and changing the following between submissions:

- ● λ, your penalty term
- ● ϵ, your convergence criterion
- ● η, your step size
- ● Fixing a random bug

(i) [0 pts] **BONUS QUESTION** (Answer this only if you have time and are confident of your other answers because this is not extra points.)

We have constructed the multiple choice problems such that every false positive will incur some negative penalty. For one of these multiple choice problems, given that there are p points, r correct answers, and k choices, what is the formula for the penalty such that the expected value of random guessing is equal to 0? (You may assume $k > r$)

$$\frac{p}{k-r}$$

Q1. [26 pts] True or False

(a) [2 pts] If the data is not linearly separable, then there is no solution to the hard-margin SVM.

● True ○ False

(b) [2 pts] Logistic Regression can be used for classification.

● True ○ False

(c) [2 pts] In logistic regression, two ways to prevent β vectors from getting too large are using a small step size and using a small regularization value.

○ True ● False

(d) [2 pts] The L2 norm is often used because it produces sparse results, as opposed to the L1 norm which does not.

○ True ● False

(e) [2 pts] For a Multivariate Gaussian, the eigenvalues of the covariance matrix are inversely proportional to the lengths of the ellipsoid axes that determine the isocontours of the density.

○ True ● False

(f) [2 pts] In a generative binary classification model where we assume the class conditionals are distributed as Poisson, and the class priors are Bernoulli, the posterior assumes a logistic form.

● True ○ False

(g) [2 pts] Maximum likelihood estimation gives us not only a point estimate, but a distribution over the parameters that we are estimating.

○ True ● False

(h) [2 pts] Penalized maximum likelihood estimators and Bayesian estimators for parameters are better used in the setting of low-dimensional data with many training examples as opposed to the setting of high-dimensional data with few training examples.

○ True ● False

(i) [2 pts] It is not a good machine learning practice to use the test set to help adjust the hyperparameters of your learning algorithm.

● True ○ False

(j) [2 pts] A symmetric positive semi-definite matrix always has nonnegative elements.

○ True ● False

(k) [2 pts] For a valid kernel function K, the corresponding feature mapping ϕ can map a finite dimensional vector into an infinite dimensional vector.

● True ○ False

(l) [2 pts] The more features that we use to represent our data, the better the learning algorithm will generalize to new data points.

○ True ● False

(m) [2 pts] A discriminative classifier explicitly models $P(Y|X)$

● True ○ False

Q2. [36 pts] Multiple Choice

(a) [3 pts] Which of the following algorithms can you use kernels with?

- ● Support Vector Machines
- ● Perceptrons
- ○ None of the above

(b) [3 pts] Cross validation:

- ● Is often used to select hyperparameters
- ○ Is guaranteed to prevent overfitting
- ○ Does nothing to prevent overfitting
- ○ None of the above

(c) [3 pts] In linear regression, L2 regularisation is equivalent to imposing a:

- ○ Logistic prior
- ● Gaussian prior
- ○ Laplace prior
- ○ Gaussian class-conditional

(d) [3 pts] Say we have two 2-dimensional Gaussian distributions representing two different classes. Which of the following conditions will result in a linear decision boundary:

- ○ Same mean for both classes
- ● Same covariance matrix for both classes
- ○ Different covariance matrix for each class
- ○ Linearly separable data

(e) [3 pts] The normal equations can be derived from:

- ● Minimizing empirical risk
- ● Assuming that $Y = \beta^T x + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$.
- ● Assuming that the $P(Y|X = x)$ is distributed normally with mean $\beta^\intercal x$ and variance σ^2
- ○ Finding a linear combination of the rows of the design matrix that minimizes the distance to our vector of labels Y

(f) [3 pts] Logistic regression can be motivated from:

- ○ Generative models with uniform class conditionals
- ● Generative models with Gaussian class conditionals
- ● Log odds being equated to an affine function of x
- ○ None of the above

(g) [3 pts] The perceptron algorithm will converge:

- ● If the data is linearly separable
- ○ Even if the data is linearly inseparable
- ○ As long as you initialize θ to all 0's
- ○ Always

(h) [3 pts] Which of the following is true:

- [x] Newton's Method typically is more expensive to calculate than gradient descent, per iteration
- [x] For quadratic equations, Newton's Method typically requires fewer iterations than gradient descent
- [ ] Gradient descent can be viewed as iteratively reweighted least squares
- [ ] None of the above

(i) [3 pts] Which of the following statements about duality and SVMs is (are) true?

- [x] Complementary slackness implies that every training point that is misclassified by a soft-margin SVM is a support vector.
- [x] When we solve the SVM with the dual problem, we need only the dot product of x_i, x_j for all i, j, and no other information about the x_i.
- [x] We use Lagrange multipliers in an optimization problem with inequality ($\leq$) constraints.
- [ ] None of the above

(j) [3 pts] Which of the following distance metrics can be computed exclusively with inner products, assuming $\Phi(x)$ and $\Phi(y)$ are feature mappings of x and y, respectively?

- [ ] $\Phi(x) - \Phi(y)$
- [x] $\|\Phi(x) - \Phi(y)\|_2^2$.
- [ ] $\|\Phi(x) - \Phi(y)\|_1$
- [ ] None of the above

(k) [3 pts] Strong duality holds for:

- [x] Hard Margin SVM
- [ ] Constrained optimization problems in general
- [x] Soft Margin SVM
- [ ] None of the above

(l) [3 pts] Which the following facts about the 'C' in SVMs is (are) true?

- [ ] As C approaches 0, the soft margin SVM is equal to the hard margin SVM
- [ ] A larger C tends to create a larger margin
- [x] None of the above
- [ ] C can be negative, as long as each of the slack variables are nonnegative

Notation:

X: the training data matrix of dimension (N, d), of N rows representing samples and d columns representing features.

x: an input data vector of dimension (1, d) of components x_i, i=1:d.

$\mathbf{x}^k$: a training example of dimension (1, d) is a row of X, k=1:N.

w: weight vector of a linear model of dimension (1, d) such that

$f(\mathbf{x}) = \mathbf{w}\,\mathbf{x}^T = \mathbf{x}\,\mathbf{w}^T = \sum_{i=1:d} w_i\, x_i$

y: target vector of dimension (N, 1) of components y^k.

α: weight vector of dimension (N, 1) of kernel method $f(\mathbf{x}) = \sum_{k=1:N} \alpha_k\, k(\mathbf{x}, \mathbf{x}^k)$

$k(\mathbf{u}, \mathbf{v})$: a kernel function (a similarity measure between two samples **u** and **v**).

True/False (36 points):

1. Stochastic gradient descent performs less computation per update than batch gradient descent. *

   TRUE ● FALSE ○

2. A function is convex if its Hessian is negative semidefinite. *

   TRUE ○ FALSE ●

3. If N < d, the solution to $X\mathbf{w}^T = \mathbf{y}$ is unique. **

   TRUE ○ FALSE ●

4. A support vector machine computes $P(y|\mathbf{x})$. **

   TRUE ○ FALSE ●

5. Adding a ridge to X^TX guarantees that it is invertible. *

TRUE ● FALSE ○

6. Grid search is less prone to being trapped in a local minimum than other heuristic search methods. ***

TRUE ● FALSE ○

7. The bootstrap method involves sampling without replacement. *

TRUE ○ FALSE ●

8. A non linearly-separable training set in a given feature space can always be made linearly-separable in another space.**

TRUE ● FALSE ●

9. Using the kernel trick, one can get non-linear decision boundaries using algorithms designed originally for linear models. *

TRUE ● FALSE ○

10. Logistic regression cannot be kernelized.*

TRUE ○ FALSE ●

11. Ridge regression, weight decay, and Gaussian processes use the same regularizer: $\|\mathbf{w}\|^2$. *

TRUE ● FALSE ○

12. Hebb's rule computes the centroid method solution if the target values are $+1/N_1$ and $-1/N_0$ (N_1 and N_0 are the number of examples of each class)**

TRUE ● FALSE ○

13. Any kernel method can be thought of as a parametric method in a possibly infinite dimensional space.*

TRUE ● FALSE ○

14. Nearest neighbors is a parametric method.*

TRUE ○ FALSE ●

15. A symmetric matrix is positive semidefinite if all its eigenvalues are positive or zero. **

TRUE ● FALSE ○

16. Zero correlation between any two random variables implies that the two random variables are independent. ***

TRUE FALSE

17. The Linear Discriminant Analysis (LDA) classifier computes the direction maximizing the ratio of between-class variance over within-class variance. ***

TRUE ● FALSE ○

18. If we repeat an experiment twice and get p-values p1 and p2, the minimum of the two p-values is the p-value of the overall experiment. ***

TRUE ○ FALSE ●

Multiple choice questions (30 points)

1. You trained a binary classifier model which gives very high accuracy on the training data, but much lower accuracy on validation data. The following may be true: *

   - [x] This is an instance of overfitting.
   - [ ] This is an instance of underfitting.
   - [x] The training was not well regularized.
   - [x] The training and testing examples are sampled from different distributions.

2. Okham in the 14th century is credited to have stated that one should “shave off unnecessary parameters of a model”. Which of the following implement that principle: **

   - [x] Regularization.
   - [ ] Maximum likelihood estimation.
   - [x] Shrinkage.
   - [ ] Empirical risk minimization.
   - [x] Feature selection.

3. Good practices to avoid overfitting include: **

   - [x] Using a two part cost function which includes a regularizer to penalize model complexity.
   - [ ] Using a good optimizer to minimize error on training data.
   - [x] Building a structure of nested subsets of models and train learning machines in each subset, starting from the inner subset, and stopping when the cross-validation error starts increasing.
   - [ ] Discarding 50% of randomly chosen samples.

4. Wrapper methods are hyper-parameter selection methods that:**

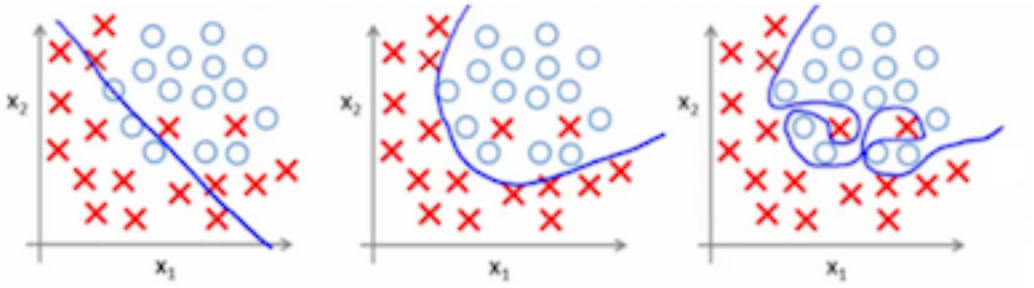
   ○ Should be used whenever possible because they are computationally efficient.

   ○ Should be avoided unless there are no other options because they are always prone to overfitting.

   ● Are useful mainly when the learning machines are “black boxes”.

   ○ Should be avoided altogether.

5. Three different classifiers are trained on the same data. Their decision boundaries are shown below. Which of the following statements are true?



   ● The leftmost classifier has high robustness, poor fit.

   ○ The leftmost classifier has poor robustness, high fit.

   ● The rightmost classifier has poor robustness, high fit.

   ○ The rightmost classifier has high robustness, poor fit.

6. What are support vectors: ***

   ○ The examples farthest from the decision boundary.

   ● The only examples necessary to compute f(**x**) in an SVM.

   ○ The class centroids.

   ● All the examples that have a non-zero weight α_k in a SVM.

7. Which of the following can only be used when training data are linearly-separable? *

- (●) Linear hard-margin SVM.
- ( ) Linear Logistic Regression.
- ( ) Linear Soft margin SVM.
- ( ) The centroid method.
- ( ) Parzen windows.

8. The number of test examples needed to get statistically significant results should be: ***

- ( ) Larger if the error rate is larger.
- (●) Larger if the error rate is smaller.
- ( ) It does not matter.

Three descriptive problems

Problem I: Gradient descent (15 points).

Given N training data points $\{(\mathbf{x}^k, y^k)\}$, k=1:N, $\mathbf{x}^k$ in R^d, and labels in y^k in {-1,1}, we seek a linear discriminant function $f(\mathbf{x}) = \mathbf{w}.\mathbf{x}$ optimizing the loss function $L(z) = e^{-z}$, for $z=y\, f(\mathbf{x})$.

Question I.1 (3 points) Is L(z) a large margin loss function? Justify your answer (a graphical justification may be useful).

Answer: Yes. This is because the loss penalizes even examples that are well classified, but the penalty decreases as you go away from the decision boundary.

Q1. [60 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [3 pts] For a *nonconvex* cost function J, which of the following step sizes guarantee that batch gradient descent will converge to the global optimum? Let i denote the ith iteration.

- ○ $\epsilon = 10^{-2}$
- ○ $\epsilon = \frac{1}{\nabla^2 J}$
- ○ $\epsilon = 10^{-i}$
- ● None of the above

(b) [3 pts] Which of the following optimization algorithms attains the optimum of an unconstrained, quadratic, convex cost function in the fewest iterations?

- ○ Batch gradient descent
- ● Newton's method
- ○ Stochastic gradient descent
- ○ The simplex method

(c) [3 pts] You train a linear classifier on 10,000 training points and discover that the training accuracy is only 67%. Which of the following, done in isolation, has a good chance of improving your training accuracy?

- ● Add novel features
- ○ Use linear regression
- ○ Train on more data
- ● Train on less data

(d) [3 pts] You train a classifier on 10,000 training points and obtain a training accuracy of 99%. However, when you submit to Kaggle, your accuracy is only 67%. Which of the following, done in isolation, has a good chance of improving your performance on Kaggle?

- ○ Set your regularization value (λ) to 0
- ● Use validation to tune your hyperparameters
- ● Train on more data
- ○ Train on less data

(e) [3 pts] You are trying to improve your Kaggle score for the spam dataset, but you must use logistic regression with no regularization. So, you decide to extract some additional features from the emails, but you forget to normalize your new features. You find that your Kaggle score goes down. Why might this happen?

- ○ The new features make the sample points linearly separable
- ● The new features have significantly more noise and larger variances than the old features
- ● The new features are uncorrelated with the emails being HAM or SPAM
- ● The new features are linear combinations of the old features

(f) [3 pts] In a soft-margin support vector machine, if we increase C, which of the following are likely to happen?

- ○ The margin will grow wider
- ● Most nonzero slack variables will shrink
- ○ There will be more points inside the margin
- ● The norm $|w|$ will grow larger

(g) [3 pts] If a hard-margin support vector machine tries to minimize $|w|^2$ subject to $y_i(X_i \cdot w + \alpha) \geq 2$ instead, what will be the width of the slab (the point-free region bracketing the decision boundary)?

- ○ $\frac{1}{\|w\|}$
- ● $\frac{4}{\|w\|}$
- ○ $\frac{2}{\|w\|}$
- ○ $\frac{1}{2\|w\|}$

(h) [3 pts] There is a 50% chance of rain on Saturday and a 30% chance of rain on Sunday. However, it is twice as likely to rain on Sunday if it rains on Saturday than if it does not rain on Saturday. What is the probability it rains on neither of the days?

- ○ 15%
- ● 40%
- ○ 25%
- ○ 45%

(i) [3 pts] The Bayes risk for a decision problem is zero when

- ○ the training data is linearly separable after lifting it to a higher-dimensional space.
- ○ the Bayes decision rule perfectly classifies the training data.
- ● the class distributions $P(X|Y)$ do not overlap.
- ● the prior probability for one class is 1.

(j) [3 pts] Consider using a Bayes decision rule classifier in a preliminary screen for cancer patients, as in Lecture 6. We want to reduce the probability that someone is classified as cancer-free when they do, in fact, have cancer. On the ROC curve for the classifier, an asymmetric loss function that implements this strategy

- ● Picks a point on the curve with higher *sensitivity* than the 0-1 loss function.
- ○ Picks a point on the curve that's closer to the y-axis than the 0-1 loss function.
- ○ Picks a point on the curve with higher *specificity* than the 0-1 loss function.
- ● Picks a point on the curve that's further from the x-axis than the 0-1 loss function.

(k) [3 pts] For which of the following cases are the scalar random variables X_1 and X_2 guaranteed to be independent?

- ○ $X_1 \sim \mathcal{N}(0,1)$ and $X_2 \sim \mathcal{N}(0,1)$.
- ○ $\mathbb{E}\left[(X_1 - \mathbb{E}[X_1])(X_2 - \mathbb{E}[X_2])\right] = -1$
- ● $\mathrm{Cov}(X_1, X_2) = 0$ and $[X_1 \quad X_2]^\top$ has a multivariate normal distribution.
- ● $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix}\right)$

(l) [3 pts] Given $X \sim \mathcal{N}(0, \Sigma)$ where the precision matrix Σ^{-1} has eigenvalues λ_i for $i = 1, \ldots, d$, the isocontours of the probability density function for X are ellipsoids whose relative axis lengths are

- ○ λ_i
- ○ $\sqrt{\lambda_i}$
- ○ $1/\lambda_i$
- ● $1/\sqrt{\lambda_i}$

(m) [3 pts] In LDA/QDA, what are the effects of modifying the sample covariance matrix as $\tilde{\Sigma} = (1 - \lambda)\Sigma + \lambda I$, where $0 < \lambda < 1$?

- ● $\tilde{\Sigma}$ is positive definite
- ● $\tilde{\Sigma}$ is invertible
- ○ Increases the eigenvalues of Σ by λ
- ● The isocontours of the quadratic form of $\tilde{\Sigma}$ are closer to spherical

(n) [3 pts] Let w^* be the solution you obtain in standard least-squares linear regression. What solution do you obtain if you scale all the input features (but not the labels y) by a factor of c before doing the regression?

- [x] $\frac{1}{c}w^*$
- [ ] cw^*
- [ ] $\frac{1}{c^2}w^*$
- [ ] c^2w^*

(o) [3 pts] In least-squares linear regression, adding a regularization term can

- [x] increase training error.
- [x] increase validation error.
- [ ] decrease training error.
- [x] decrease validation error.

(p) [3 pts] You have a design matrix $X \in \mathbb{R}^{n \times d}$ with $d = 100{,}000$ features and and vector $y \in \mathbb{R}^n$ of binary 0-1 labels. When you fit a logistic regression model to your design matrix, your test error is much worse than your training error. You suspect that many of the features are useless and are therefore causing overfitting. What are some ways to eliminate the useless features?

- [x] Use ℓ_1 regularization.
- [ ] Use ℓ_2 regularization.
- [x] Iterate over features; check if removing feature i increases validation error; remove it if not.
- [ ] If the ith eigenvalue λ_i of the sample covariance matrix is 0, remove the ith feature/column.

(q) [3 pts] Recall the data model, $y_i = f(X_i) + \epsilon_i$, that justifies the least-squares cost function in regression. The statistical assumptions of this model are, for all i,

- [x] ϵ_i comes from a Gaussian distribution.
- [x] all ϵ_i have the same mean
- [ ] all y_i have the same mean
- [x] all y_i have the same variance

(r) [3 pts] How does ridge regression compare to linear regression with respect to the bias-variance tradeoff?

- [x] Ridge regression usually has higher bias.
- [ ] Ridge regression usually has higher variance.
- [ ] Ridge regression usually has higher irreducible error.
- [x] Ridge regression's variance approaches zero as the regularization parameter $\lambda \to \infty$.

(s) [3 pts] Which of the following quantities affect the bias-variance tradeoff?

- [x] λ, the regularization coefficient in ridge regression
- [ ] ϵ, the learning rate in gradient descent
- [x] C, the slack parameter in soft-margin SVM
- [x] d, the polynomial degree in least-squares regression

(t) [3 pts] Which of the following statements about maximum likelihood estimation are true?

- [x] MLE, applied to estimate the mean parameter μ of a normal distribution $\mathcal{N}(\mu, \Sigma)$ with a known covariance matrix Σ, returns the mean of the sample points
- [ ] For a sample drawn from a normal distribution, the likelihood $\mathcal{L}(\mu, \sigma; X_1, \ldots, X_n)$ is equal to the probability of drawing exactly the points $X_1, \ldots, X_n$ (in that order) when you draw n random points from $\mathcal{N}(\mu, \sigma)$
- [ ] MLE, applied to estimate the covariance parameter Σ of a normal distribution $\mathcal{N}(\mu, \Sigma)$, returns $\hat{\Sigma} = \frac{1}{n}X^TX$, where X is the design matrix
- [x] Maximizing the log likelihood is equivalent to maximizing the likelihood

Q1. [60 pts] Multiple Choice

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [3 pts] Which of the following learning algorithms will return a classifier if the training data is not linearly separable?

- ○ Hard-margin SVM
- ● Soft-margin SVM
- ○ Perceptron
- ● Linear Discriminant Analysis (LDA)

(b) [3 pts] With a soft-margin SVM, which samples will have non-zero slack variables ξ_i?

- ● All misclassified samples
- ● All samples inside the margin
- ○ All samples lying on the margin boundary
- ○ All samples outside the margin

(c) [3 pts] Recall the soft-margin SVM objective function $|\mathbf{w}|^2 + C\sum_i \xi_i$. Which value of C is most likely to overfit the training data?

- ○ $C = 0.01$
- ○ $C = 1$
- ○ $C = 0.00001$
- ● $C = 100$

(d) [3 pts] There are several ways to formulate the hard-margin SVM. Consider a formulation in which we try to directly maximize the margin β. The training samples are $X_1, X_2, \ldots, X_n$ and their labels are $y_1, y_2, \ldots, y_n$. Which constraints should we impose to get a correct SVM? (*Hint*: Recall the formula for the distance from a point to a hyperplane.) Maximize β subject to ...

- ○ $y_i X_i^T \mathbf{w} \leq \beta \quad \forall i \in [1, n]$.
- ● $y_i X_i^T \mathbf{w} \geq \beta \quad \forall i \in [1, n]$.
- ○ $|\mathbf{w}| \geq 1$.
- ● $|\mathbf{w}| = 1$.

(e) [3 pts] In the homework, you trained classifiers on the digits dataset. The features were the pixels in each image. What features could you add that would improve the performance of your classifier?

- ○ Maximum pixel intensity
- ● Average pixel intensity
- ● Number of enclosed regions
- ● Presence of a long horizontal line

(f) [3 pts] The Bayes risk for a decision problem is zero when

- ● the class distributions $P(X|Y)$ do not overlap.
- ○ the training data is linearly separable.
- ○ the loss function $L(z, y)$ is symmetrical.
- ○ the Bayes decision rule perfectly classifies the training data.

(g) [3 pts] Let $L(z, y)$ be a loss function (where y is the true class and z is the predicted class). Which of the following loss functions will *always* lead to the same Bayes decision rule as L?

- ● $L_1(z, y) = aL(z, y)$, $a > 0$
- ○ $L_2(z, y) = aL(z, y)$, $a < 0$
- ● $L_3(z, y) = L(z, y) + b$, $b > 0$
- ● $L_4(z, y) = L(z, y) + b$, $b < 0$

(h) [3 pts] Gaussian discriminant analysis

- ○ models $P(Y = y|X)$ as a Gaussian.
- ● models $P(Y = y|X)$ as a logistic function.
- ● is an example of a generative model.
- ● can be used to classify points without ever computing an exponential.

(i) [3 pts] Which of the following are valid covariance matrices?

- ○ $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$
- ● $B = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$
- ○ $C = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$
- ● $D = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

(j) [3 pts] Consider a d-dimensional multivariate normal distribution that is isotropic (i.e., its isosurfaces are spheres). Let Σ be its $d \times d$ covariance matrix. Let I be the $d \times d$ identity matrix. Let σ be the standard deviation of any one component (feature). Then

- ○ $\Sigma = \sigma I$.
- ● $\Sigma = \sigma^2 I$.
- ○ None of the above.
- ○ $\Sigma = \frac{1}{\sigma} I$.
- ○ $\Sigma = \frac{1}{\sigma^2} I$.

(k) [3 pts] In least-squares linear regression, imposing a Gaussian prior on the weights is equivalent to

- ○ logistic regression
- ○ adding a Laplace-distributed penalty term
- ● L_2 regularization
- ○ L_1 regularization

(l) [3 pts] Logistic regression

- ○ assumes that we impose a Gaussian prior on the weights.
- ● minimizes a convex cost function.
- ○ has a closed-form solution.
- ● can be used with a polynomial kernel.

(m) [3 pts] Ridge regression

- ○ is more sensitive to outliers than ordinary least-squares.
- ● reduces variance at the expense of higher bias.
- ○ adds an L_1-norm penalty to the cost function.
- ○ often sets several of the weights to zero.

(n) [3 pts] Given a design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and labels $\mathbf{y} \in \mathbb{R}^n$, which of the following techniques could potentially decrease the empirical risk on the training data (assuming the loss is the squared error)?

- ● Adding the feature "1" to each data point.
- ● Adding polynomial features to each data point.
- ● Centering the vector $\mathbf{y}$ by subtracting the mean $\bar{y}$ from each component y_i.
- ○ Penalizing the model weights with L_2 regularization.

(o) [3 pts] In terms of the bias-variance trade-off, which of the following is/are substantially more harmful to the test error than the training error?

- ◯ Bias
- ◯ Loss
- ● Variance
- ◯ Risk

(p) [3 pts] Consider the bias-variance trade-off in fitting least-squares surfaces to two data sets. The first is US census data, in which we want to estimate household income from the other variables. The second is synthetic data we generated by writing a program that randomly creates samples from a known normal distribution, and assigns them y-values on a known smooth surface $y = f(\mathbf{x})$ plus noise drawn from a known normal distribution. We can compute or estimate with high accuracy

- ◯ the bias component of the empirical risk for the US census data.
- ● the bias component of the empirical risk for the synthetic data.
- ◯ the variance component of the empirical risk for the US census data.
- ● the variance component of the empirical risk for the synthetic data.

(q) [3 pts] The kernel trick

- ◯ is necessary if we want to add polynomial features to a learning algorithm.
- ● can improve the speed of high-degree polynomial regression.
- ◯ can be applied to any learning algorithm.
- ◯ can improve the speed of learning algorithms when the number of samples is very large.

(r) [3 pts] In the usual formulation of soft-margin SVMs, each training sample has a slack variable $\xi_i \geq 0$ and we impose a regularization cost $C \sum_i \xi_i$. Consider an alternative formulation where we impose the additional constraints $\xi_i = \xi_j$ for all i, j. How does the minimum objective value $|\mathbf{w}|^2 + C \sum_i \xi_i$ obtained by the new method compare to the one obtained by the original soft-margin SVM?

- ◯ They are always equal.
- ◯ Original SVM minimum $\geq$ new minimum.
- ● New minimum $\geq$ original SVM minimum.
- ◯ New minimum is sometimes larger and sometimes smaller.

(s) [3 pts] In Gaussian discriminant analysis, if two classes come from Gaussian distributions that have different means, may or may not have different covariance matrices, and may or may not have different priors, which decision boundary shapes are possible?

- ● a hyperplane
- ◯ a surface that is not a quadric
- ● a nonlinear quadric surface (quadric = the isosurface of a quadratic function)
- ● the empty set (the classifier always returns the same class)

(t) [3 pts] Let the class conditionals be given by $P(X|Y = i) \sim \mathcal{N}(\mathbf{0}, \Sigma_i)$, where $i \in \{0, 1\}$ and $\Sigma_0 = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ and $\Sigma_1 = \begin{bmatrix} b & 0 \\ 0 & a \end{bmatrix}$ with $a, b > 0$, $a \neq b$. Both conditionals have mean zero, and both classes have the prior probability $P(Y = 0) = P(Y = 1) = 0.5$. What is the shape of the decision boundary?

- ◯ a line
- ● multiple lines
- ◯ a nonlinear quadratic curve
- ◯ not defined

Q1. [60 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [3 pts] Let A be a real, symmetric $n \times n$ matrix. Which of the following are true about A's eigenvectors and eigenvalues?

- ● A can have no more than n distinct eigenvalues
- ○ The vector $\vec{0}$ is an eigenvector, because $A\vec{0} = \lambda\vec{0}$
- ○ A can have no more than $2n$ distinct unit-length eigenvectors
- ● We can find n mutually orthogonal eigenvectors of A

There can be infinitely many unit-length eigenvectors if the multiplicity of any eigenvector is greater than 1 (so the eiganspace is a plane, and you can pick any vector on the unit circle on that plane).

The 0 vector is not an eigenvector by definition.

(b) [3 pts] The matrix that has eigenvector $[1, 2]^\top$ with eigenvalue 2 and eigenvector $[-2, 1]^\top$ with eigenvalue 1 (note that these are not unit eigenvectors!) is

- ○ $\begin{bmatrix} 9 & -2 \\ -2 & 6 \end{bmatrix}$
- ○ $\begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix}$
- ○ $\begin{bmatrix} 9/5 & -2/5 \\ -2/5 & 6/5 \end{bmatrix}$
- ● $\begin{bmatrix} 6/5 & 2/5 \\ 2/5 & 9/5 \end{bmatrix}$

(c) [3 pts] Consider a **binary classification** problem where we know both of the class conditional distributions exactly. To compute the risk,

- ○ we need to know all the sample points
- ● we need to know the class prior probabilities
- ● we need to know the loss function
- ○ we need to use gradient descent

(d) [3 pts] Assuming we can find algorithms to minimize them, which of the following cost functions will encourage **sparse solutions** (i.e., solutions where many components of w are zero)?

- ● $\|Xw - y\|_2^2 + \lambda\|w\|_1$
- ● $\|Xw - y\|_2^2 + \lambda \cdot (\#$ of nonzero components of $w)$
- ● $\|Xw - y\|_2^2 + \lambda\|w\|_1^2$
- ○ $\|Xw - y\|_2^2 + \lambda\|w\|_2^2$

The first answer is Lasso, which we know finds sparse solutions. The second is Lasso with the penalty squared. Squaring this will leave the same isocontours and this will keep the same properties as Lasso. The third cost function penalizes solutions that are not sparse and will naturally encourage sparse solutions. The last solution is ridge regression, which shrinks weights but does not set weights to zero.

(e) [3 pts] Which of the following statements about **logistic regression** are correct?

- ● The cost function of logistic regression is convex
- ○ The cost function of logistic regression is concave
- ○ Logistic regression uses the squared error as the loss function
- ○ Logistic regression assumes that each class's points are generated from a Gaussian distribution

(f) [3 pts] Which of the following statements about stochastic gradient descent and Newton's method are correct?

● Newton's method often converges faster than stochastic gradient descent, especially when the dimension is small

● Newton's method converges in one iteration when the cost function is exactly quadratic with one unique minimum.

○ If the function is continuous with continuous derivatives, Newton's method always finds a local minimum

○ Stochastic gradient descent reduces the cost function at every iteration.

(g) [3 pts] Let $X \in \mathbb{R}^{n \times d}$ be a design matrix containing n sample points with d features each. Let $y \in \mathbb{R}^n$ be the corresponding real-valued labels. What is always true about **every** solution w^* that **locally** minimizes the **linear least squares** objective function $\|Xw - y\|_2^2$, no matter what the value of X is?

- ○ $w^* = X^+ y$ (where X^+ is the pseudoinverse)
- ● w^* satisfies the normal equations
- ○ w^* is in the null space of X
- ● All of the local minima are global minima

Top left: $w^* = X^+ y$ is the least squares solution with **least norm**. If the null-space of X is non-empty, there are infinitely many other solutions that minimize $\|Xw - y\|_2^2$ but that have larger norm.

Bottom left: If w^* was in the null space of X, then $Xw^* = 0$. This can only be a solution to the linear least squares objective if and only if y is also in the null-space of X.

Top right: The normal equations $A^T Aw = A^T y$ define all values of w that make zero the gradient of the least squares objective. Therefore any minimizer must satisfy them.

Bottom right: The objective is convex, and therefore all local minimizers are also global minimizers.

(h) [3 pts] We are using **linear discriminant analysis** to classify points $x \in \mathbb{R}^d$ into **three** different classes. Let S be the set of points in $\mathbb{R}^d$ that our trained model classifies as belonging to the first class. Which of the following are true?

- ○ The decision boundary of S is always a hyperplane
- ● S can be the whole space $\mathbb{R}^d$
- ● The decision boundary of S is always a subset of a union of hyperplanes
- ● S is always connected (that is, every pair of points in S is connected by a path in S)

Top left: Given that we have three classes, S is defined by two linear inequalities, and therefore its boundary may not be a hyperplane.

Bottom left: Given that S is defined as the points satisfying a set of inequalities, its boundary is a subset of the hyperplanes defined by each of the linear inequalities.

Top right: If the prior for the first class is high enough, the probability of that class could be higher everywhere, and hence S would be the whole space. For example, take $\mu_1 = \mu_2 = \mu_3$ and $\pi_1 > \pi_2 = \pi_3$.

Bottom right: S is a convex polytope defined by the intersection of half-spaces (i.e. the points satisfying a set of linear inequalities). This is a convex set, and therefore it is connected.

(i) [3 pts] Which of the following apply to **linear discriminant analysis**?

- ● You calculate the sample mean for each class
- ● It approximates the Bayes decision rule
- ○ You calculate the sample covariance matrix using the mean of all the data points
- ○ The model produced by LDA is never the same as the model produced by QDA

Top left: Calculating the sample mean within each class is part of LDA by definition

Bottom left: You calculate the sample covariance using the mean for each class, not the mean of all the data points

Top right: LDA finds what the Bayes decision rule would be under the assumption the class conditionals have normal distributions, parameterized by the sample means and covariance

Bottom right: QDA can produce the same covariance for each class as LDA

(j) [3 pts] Which of the following are reasons why you might adjust your model in ways that increase the bias?

- ○ You observe high training error and high validation error
- ● You observe low training error and high validation error
- ● You have few data points
- ○ Your data are not linearly separable

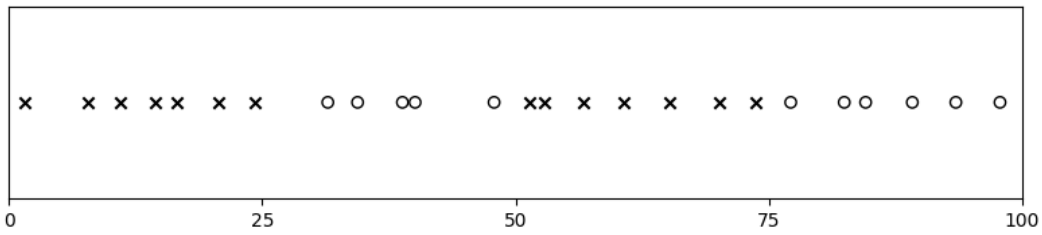
Top left: High training and validation error is a sign of underfitting; higher bias leads to greater underfitting hence you would not want the bias to increase

Bottom left: With few data points, noise in the data has a larger effect on the model. Methods to reduce overfitting to the noise would increase the bias

Top right: Low training error and high validation error is a sign of overfitting; methods to decrease overfitting increase bias

Bottom right: If the data are not linearly separable, you need a more complex decision boundary. Higher bias would reduce the complexity of the decision boundary.

(k) [3 pts] Suppose you are given the one-dimensional data $\{x_1, x_2, \ldots, x_{25}\}$ illustrated below and you have only a **hard-margin support vector machine** (with a fictitious dimension) at your disposal. Which of the following modifications can give you 100% training accuracy?



○ Centering the data

● Add a feature that is 1 if $x \leq 50$, or -1 if $x > 50$

○ The performance of SVM is shift invariant, so centering the data won't affect the result;

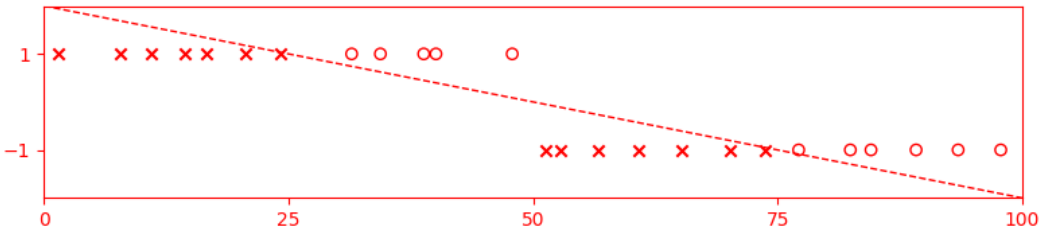
● See image below;

○ Add a feature x_i^2

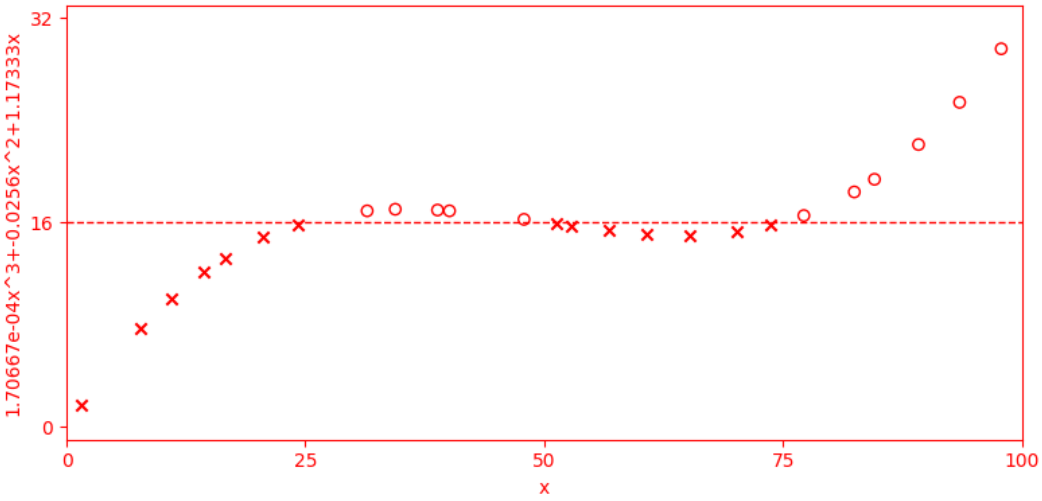
● Add two features, x_i^2 and x_i^3

○ A line can separate a quadratic function into at most 3 segments and is not sufficient;

● A line can separate a cubic function into 4 segments. See image below.



Adding "1 if $x_i \leq 50$..." feature



Adding x_i^2 and x_i^3

(l) [3 pts] You are performing **least-squares polynomial regression**. As the degree of your polynomials increases, which of the following is commonly seen to go down at first but then go up?

- ○ Training error
- ● Validation error
- ○ Variance
- ○ Bias

(m) [3 pts] Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous, smooth function whose derivative $f'(x)$ is also continuous. Suppose f has a unique global minimum $x^* \in (-\infty, \infty)$, and you are using **gradient descent** to find x^*. You fix some $x^{(0)} \in \mathbb{R}$ and $\epsilon > 0$, and run $x^{(t)} = x^{(t-1)} - \epsilon f'(x^{(t-1)})$ repeatedly. Which of the following statements are true?

- ○ Gradient descent is sure to converge, to *some* value, for any step size $\epsilon > 0$
- ○ Assuming gradient descent converges, it converges to x^* if and only if f is convex
- ● If f has a local minimum x' different from the global one, i.e., $x' \neq x^*$, and $x^{(t)} = x'$ for some t, gradient descent will not converge to x^*
- ● If, additionally, f is the objective function of logistic regression, and gradient descent converges, then it converges to x^*

The top-left option is false because for a large enough step size, gradient descent may not converge. The bottom-left option is correct because $f'(x') = 0$, so gradient descent will never move from a local minimium. The top-right option is false because you could "accidentally" initialize GD at x^* even if f is non-convex. The bottom-right option is correct because the objective of logistic regression is convex.

(n) [3 pts] Suppose you are trying to choose a good subset of features for a **least-squares linear regression** model. Let algorithm A be forward stepwise selection, where we start with zero features and at each step add the new feature that most decreases validation error, stopping only when validation error starts increasing. Let algorithm B be similar, but at each step we include the new feature that most decreases **training** error (measured by the usual cost function, mean squared error), stopping only when training error starts increasing. Which of the following is true?

- ○ Algorithm B will select no more features than Algorithm A does
- ○ The first feature chosen by the two algorithms will be the same
- ● Algorithm B will select at least as many features as Algorithm A does
- ○ Algorithm A sometimes selects features that Algorithm B does not select

Algorithm B will always select all features, since training error cannot decrease with the addition of a new feature in a linear regression model. There is no guarantee that the first feature chosen will be the same.

(o) [3 pts] Suppose you have a **multivariate normal distribution** with a positive definite covariance matrix Σ. Consider a second multivariate Gaussian distribution whose covariance matrix is $\kappa\Sigma$, where $\kappa = \cos\theta > 0$. Which of the following statements are true about the ellipsoidal isocontours of the second distribution, compared to the first distribution?

- ○ The principal axes of the ellipsoids would be rotated by θ
- ○ The principal axes (radii) of the ellipsoids will be scaled by $1/\kappa$
- ○ The principal axes (radii) of the ellipsoids will be scaled by κ
- ● The principal axes (radii) of the ellipsoids will be scaled by $\sqrt{\kappa}$

Multiplying Σ by κ multiplies the eigenvalues by κ. The axes of the ellipsoids are scaled by the square roots of the eigenvalues of Σ.

(p) [3 pts] Suppose M and N are **positive semidefinite** matrices. Under what conditions is $M - N$ certain to be positive semidefinite?

- ○ Never
- ○ If M and N share all the same eigenvectors
- ● The smallest eigenvalue of M is greater than the largest eigenvalue of N
- ○ The largest eigenvalue of M is greater than the largest eigenvalue of N

We know that $v^T M v \geq \lambda_{\min}(M)$ for all v and $w^T N w \leq \lambda_{\max}(N)$ for all w. Thus if $\lambda_{\min}(M) \geq \lambda_{\max}(N)$,

$$v^T M v \geq \lambda_{\min}(M) \geq \lambda_{\max}(N) \geq v^T N v$$
$$v^T (M - N) v \geq 0.$$

These do not require the same eigenvectors. For the last case, consider

$$M = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}, \; N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \; M - N = \begin{bmatrix} 1 & 0 \\ 0 & -0.5 \end{bmatrix}.$$

$M - N$ is not positive semidefinite since it has an eigenvalue of -0.5, but $\lambda_{\max}(M) > \lambda_{\min}(N)$.

(q) [3 pts] You are given four sample points $X_1 = [-1, -1]^\top$, $X_2 = [-1, 1]^\top$, $X_3 = [1, -1]^\top$, and $X_4 = [1, 1]^\top$. Each of them is in class C or class D. For what feature representations are the lifted points $\Phi(X_i)$ *guaranteed* to be **linearly separable** (with no point lying exactly on the decision boundary) for every possible class labeling?

- ○ $\Phi(x) = [x_1, x_2, 1]$
- ○ $\Phi(x) = [x_1, x_2, x_1^2 + x_2^2, 1]$
- ○ $\Phi(x) = [x_1^2, x_2^2, x_1, x_2, 1]$
- ● $\Phi(x) = [x_1^2, x_2^2, x_1 x_2, x_1, x_2, 1]$

Consider the labels $y_1 = 1, y_2 = 0, y_3 = 0, y_4 = 1$. No linear classifier (with or without bias) can classify these samples. Also, $x_1^2 = x_2^2 = 1$ for all X_i, and $x_1^2 + x_2^2 = 2$ for all X_i.

The last option is a quadratic kernel, which can learn e.g. an elliptical decision boundary in the original sample space to perfectly classify the above labeling.

(r) [3 pts] Let $L_i(w)$ be the loss corresponding to a sample point X_i with label y_i. The update rule for **stochastic gradient descent** with step size ϵ is

- ○ $w_{\text{new}} \leftarrow w - \epsilon \nabla_{X_i} L_i(w)$
- ○ $w_{\text{new}} \leftarrow w - \epsilon \sum_{i=1}^n \nabla_{X_i} L_i(w)$
- ● $w_{\text{new}} \leftarrow w - \epsilon \nabla_w L_i(w)$
- ○ $w_{\text{new}} \leftarrow w - \epsilon \sum_{i=1}^n \nabla_w L_i(w)$

Any option taking the gradient w.r.t. X_i is an incorrect update rule for the weights w, so that leaves only the solutions which use ∇_w. Any option which sums over multiple gradients is a batch method, which leaves only one option, the solution.

(s) [3 pts] Suppose you have a sample in which each point has d features and comes from class C or class D. The class conditional distributions are $(X_i | y_i = C) \sim N(\mu_C, \sigma_C^2)$ and $(X_i | y_i = D) \sim N(\mu_D, \sigma_D^2)$ for unknown values $\mu_C, \mu_D \in \mathbb{R}^d$ and $\sigma_C^2, \sigma_D^2 \in \mathbb{R}$. The class priors are π_C and π_D. We use 0-1 loss.

- ● If $\pi_C = \pi_D$ and $\sigma_C = \sigma_D$, then the Bayes decision rule assigns a test point z to the class whose mean is closest to z.
- ● If $\pi_C = \pi_D$, then the Bayes decision rule is $r^*(z) = \text{argmin}_{A \in \{C,D\}} \left(|z - \mu_A|^2 / (2\sigma_A^2) + d \ln \sigma_A \right)$
- ● If $\sigma_C = \sigma_D$, then the Bayes decision boundary is always linear.
- ○ If $\sigma_C = \sigma_D$, then QDA will always produce a linear decision boundary when you fit it to your sample.

(t) [3 pts] Let $f \in [0, 1]$ be the *unknown*, fixed probability that a person in a certain population owns a dog (how cute!). We model f with a hypothesis $h \in [0, 1]$. Before we observe any data at all, we can't even guess what f might be, so we set our prior probability for f to be the uniform distribution, i.e., $P(f = h) = 1$ over $h \in [0, 1]$. Now, we pick one person from the population, and it turns out that they have a cute little labradoodle named Dr. Frankenstein. Which of the following is true about the posterior probability that $f = h$ given this one sample point?

- ○ The posterior is uniform over $h \in [0, 1]$.
- ● The posterior increases linearly over $h \in [0, 1]$.
- ○ The posterior increases nonlinearly over $h \in [0, 1]$.
- ○ The posterior is a delta function at 1.

Q1. [40 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [4 pts] Let X be an $m \times n$ matrix. Which of the following are always equal to rank(X)?

- ● A: rank(X^T)
- ○ C: m – dimension(nullspace(X))
- ● B: rank($X^T X$)
- ● D: dimension(rowspace(X))

Option C is not equal because by Rank-Nullity Theorem: n−dim(nullspace(X)) =rank(X)
Options A and D are equal, since dim(rowspace(X)) =dim(columnspace(X^T)) =rank(X^T) =rank(X)

(b) [4 pts] Which of the following types of square matrices can have negative eigenvalues?

- ● A: a symmetric matrix
- ● C: an orthonormal matrix (M such that $M^\top M = I$)
- ○ B: $I - uu^T$ where u is a unit vector
- ● D: $\nabla^2 f(x)$ where $f(x)$ is a Gaussian PDF

Top left: A symmetric matrix can have negative eigenvalues, they just have to be real. Bottom left: u is a unit vector that can be expressed as a linear combination of the standard vectors $\sum_{i=1}^{n} c_i * e_i$ where $c_i < 1$. $(\sum_{i=1}^{n} c_i * e_i)(\sum_{i=1}^{n} c_i * e_i)^T = (\sum_{i=1}^{n} c_i * e_i)(\sum_{i=1}^{n} c_i * e_i)^T$ Top right: An orthogonal matrix has eigenvalues of 1 and -1. Bottom right: The gaussian of a PDF is a concave function, thus the hessian must have negative eigenvalues.

(c) [4 pts] Choose the correct statement(s) about Support Vector Machines (SVMs).

- ● A: if a finite set of training points from two classes is linearly separable, a **hard-margin** SVM will always find a decision boundary correctly classifying every training point
- ○ B: if a finite set of training points from two classes is linearly separable, a **soft-margin** SVM will always find a decision boundary correctly classifying every training point
- ● C: every trained two-class **hard-margin** SVM model has at least one point of each class at a distance of exactly $1/\|w\|$ (the margin width) from the decision boundary
- ○ D: every trained two-class **soft-margin** SVM model has at least one point of each class at a distance of exactly $1/\|w\|$ (the margin width) from the decision boundary

Option A is correct: fundamental material about SVMs from lectures.

(d) [4 pts] Suppose we perform least-squares linear regression, but we don't assume that all weight vectors are equally reasonable; instead, we use the maximum *a posteriori* method to impose a normally-distributed prior probability on the weights. Then we are doing

- ● A: L_2 regularization
- ○ C: logistic regression
- ○ B: Lasso regression
- ● D: ridge regression

As shown in Lecture 13, the Bayesian justification for ridge regression is derived by applying MAP to the posterior probability with a Gaussian prior on the weights.

(e) [4 pts] Which of the following statements regarding ROC curves are true?

● A: the ROC curve is monotonically increasing

○ B: for a logistic regression classifier, the ROC curve's horizontal axis is the posterior probability used as a threshold for the decision rule

○ C: the ROC curve is concave

● D: if the ROC curve passes through $(0, 1)$, the classifier is always correct (on the test data used to make the ROC curve)

The axes of an ROC curve do not correspond to the "knob" we're turning when we plot the curve.

Always predicting positive will give us 100

Does not have to be concave, just needs to be increasing.

Since it's increasing, the curve is a horizontal line at y=1. So, we have no false positives nor false negatives.

(f) [4 pts] One way to understand regularization is to ask which vectors minimize the regularization term. Consider the set of unit vectors in the plane: $\{x \in \mathbb{R}^2 : \|x\|_2^2 = 1\}$. Which of the following regularization terms are minimized solely by the four unit vectors $\{(0, 1), (1, 0), (-1, 0), (0, -1)\}$ and no other unit vector?

● A: $f(x) = \|x\|_0$ = the # of nonzero entries of x

● B: $f(x) = \|x\|_1$

○ C: $f(x) = \|x\|_2^2$

○ D: $f(x) = \|x\|_\infty = \max\{|x_1|, |x_2|\}$

The first option is almost true by definition: these are the sparsest unit vectors. The second option follows Cauchy–Schwartz. Intuitively, however, we know also that the ℓ_1-norm promotes sparsity, so we should expect this to be true. Finally, notice that $\|x\|_2^2$ always equals 1 and that $\max(x_1, x_2)$ is minimized when $x_1 = x_2$, so both these options are incorrect.

(g) [4 pts] Suppose we train a soft-margin SVM classifier on data with d-dimensional features and binary labels. Below we have written four pairs of the form "modification → effect." For which ones would a model trained on the modified data **always** have the corresponding effect relative to the original model?

● A: augment the data with polynomial features → optimal value of the objective function (on the training points) decreases or stays the same

○ B: multiply each data point by a fixed invertible $d \times d$ matrix A; i.e., $X_i \leftarrow AX_i$ → all training points are classified the same as before

● C: multiply each data point by a fixed orthonormal $d \times d$ matrix U and add a fixed vector $z \in \mathbb{R}^d$; i.e., $X_i \leftarrow UX_i + z$ → all training points are classified the same as before.

○ D: normalize each feature so that its mean is 0 and variance is 1 → all training points are classified the same as before

1) The original optimal values are possible in the modified setup, since we can use the original w on the original features and set $w_i = 0$ for all w_i corresponding to the newly added features. Thus, the new optimum must be at least as good as the old one.

2) An invertible matrix could still send the data points to a transformed space in which the relative scale of different components is very different from the original, for instance, affecting the optimization of the new w.

3) Multiplying by an orthogonal matrix is an isometry, meaning that it preserves the norm of all vectors, so the relative scale stays the same. Adding z to every data point just shifts them. Concretely, take $w' = Uw$ and $\alpha' = \alpha - z^T w'$ to recover the original optimum, since $\|w'\| = \|Uw\| = \|w\|$.

4) Normalizing each feature changes the relative scale of different components; see 2).

(h) [4 pts] A real-valued $n \times n$ matrix P is called a projection matrix if $P^2 = P$. Select all the true statements about eigenvalues of P.

- [x] A: P can have an eigenvalue of 0
- [x] B: P can have an eigenvalue of 1
- [ ] C: P can have an eigenvalue of -1
- [ ] D: P can have an eigenvalue that isn't 0, 1, or -1

(i) [4 pts] Let X be a real-valued $n \times d$ matrix. Let Ω be a diagonal, real-valued $n \times n$ matrix whose diagonal entries are all positive. Which of the following are true of the matrix product $M = X^T \Omega X$?

- [ ] A: M could have negative eigenvalues
- [x] B: M could have eigenvalues equal to zero
- [x] C: M could have positive eigenvalues
- [ ] D: the eigenvalues of M are the values on the diagonal of Ω

The matrix M is PSD:

$$\forall v \in R^d, v^T X^T \Omega X v = \sum_{i=1}^{n} \omega_{ii} (x_i^T v)^2 \geq 0$$

It therefore cannot have negative eigenvalues. It can have positive eigenvalues, or eigenvalues equal to 0 (for example, when all entries of X are 0). Note that the eigenvalues are not necessarily the values along the diagonal of Ω. That would be the case when the matrix X is orthogonal, but the matrix X isn't even necessarily square.

(j) [4 pts] Which of the following regression methods always have just one unique optimum, regardless of the data?

- [ ] A: least Squares Regression
- [x] B: ridge Regression
- [ ] C: Lasso Regression
- [ ] D: logistic Regression

Only ridge regression is guaranteed to converge to have a unique optimum as the regularization term makes the objective function strictly convex. Other options may also have convex objectives, but there may be multiple points in weight space at which the same loss value is obtained.

Q1. [40 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [4 pts] Recall that in subset selection, we attempt to identify poorly predictive features and ignore them. Which of the following are reasons why we may seek to drop features available to our model?

- ○ A: To reduce model bias
- ● B: To reduce model variance
- ● C: To increase speed of prediction on test points
- ● D: To improve model interpretability

Removing features likely increases bias, since the model becomes less expressive. On the other hand, all features increase variance; therefore, dropping features may reduce the variance of the model. Of course, having fewer features may also make prediction slightly faster and improve interpretability.

(b) [4 pts] Consider a random variable $X \sim \mathcal{N}(\mu, \Sigma) \in \mathbb{R}^d$, where the multivariate Gaussian probability density function (PDF) is axis-aligned, Σ is positive definite, and the standard deviation along coordinate axis i is σ_i. Select all that apply.

- ○ A: The d features of X are uncorrelated but not necessarily independent
- ○ B: $\Sigma = \text{diag}(\sqrt{\sigma_1}, \sqrt{\sigma_2}, \ldots, \sqrt{\sigma_d})$
- ● C: Σ has a symmetric square root $\Sigma^{1/2}$ with eigenvalues $\sigma_1, \sigma_2, \ldots, \sigma_d$
- ● D: $(X-\mu)^\top \Sigma^{-1}(X-\mu) \geq 0$

A is incorrect: It is true that because the multivariate Gaussian is axis-aligned, its components are uncorrelated. However, the uncorrelated multivariate Gaussian also implies independence here, since $p(X;\mu,\Sigma) = \prod_{i=1}^n p(x_i;\mu_i,\sigma_i^2)$ (see Option C).

B is incorrect: The eigenvectors are standard basis vectors, so Σ must be a diagonal matrix, but with elements $\text{diag}(\sigma_1^2, \ldots, \sigma_n^2)$.

C is correct: Because Σ is a diagonal matrix, expanding out the probability density function gives us option C.

D is correct: Σ is positive definite, so Σ^{-1} is also positive definite. Thus, $(X-\mu)^\top \Sigma^{-1}(X-\mu)$ must be nonnegative.

(c) [4 pts] Given a design matrix $X \in \mathbb{R}^{n \times d}$ representing n sample points with d features, you compute the sample covariance matrix M of your dataset and find that its determinant is $\det M = 0$. What do you know to be true?

- ● A: There is some direction in $\mathbb{R}^d$ along which the sample points have zero variance
- ○ B: The covariance matrix M is positive definite
- ● C: The columns of the centered design matrix $\dot{X}$ are linearly dependent
- ○ D: The rows of the centered design matrix $\dot{X}$ are linearly dependent

(d) [4 pts] For classification problems with two features ($d = 2$, test point $z \in \mathbb{R}^2$), which of the following methods have posterior probability distributions of the form $P(Y|X=z) = s(Az_1^2 + Bz_2^2 + Cz_1z_2 + Dz_1 + Ez_2 + F)$ where s is the logistic function $s(\gamma) = \frac{1}{1+e^{-\gamma}}$ and $A, B, C, D, E, F \in \mathbb{R}$ can all be nonzero?

- ○ A: Logistic regression with linear features
- ● B: Linear discriminant analysis (LDA) with quadratic features
- ● C: Logistic regression with quadratic features
- ● D: Quadratic discriminant analysis (QDA) with linear features

All but A, as standard logistic regression does not have the potential for A, B, C to be nonzero.

(e) [4 pts] Which of the following statements regarding Bayes decision theory are true?

○ A: If the Bayes optimal classifier $r^*(x)$ correctly classifies all points in the design matrix X with 100% accuracy, its Bayes risk must be zero

● B: With 0-1 loss, the two-class Bayes optimal classifier $r^*(x)$ classifies points in a way that minimizes the probability of misclassification

○ C: If you have a design matrix X and you are given the Bayes optimal classifier $r^*(x)$, then you sample a different design matrix X' from the same distribution(s), the original $r^*(x)$ is no longer optimal

○ D: None of the above

A: There may still be some nonzero probability for posteriors of non-selected classes. The optimal classifier aims to achieve lowest Expected Risk, there are no guarantees that this will always be zero.

B: This might be easier to view in terms of having 0-1 Loss. When we have 0-1 loss and decide between class C or D or Loss is effectively just the probability of the opposing class. Therefore to minimize our loss, we would choose to avoid the class with lower probability. This results in ultimately the lowest misclassification error in expectation.

C: The Bayes optimal classifier has nothing to do with any particular set of sample points drawn from the underlying distributions.

(f) [4 pts] We want to minimize the function

$$f(\beta) = \begin{cases} \beta^2 & \text{if } \beta \leq 0, \\ \beta & \text{otherwise.} \end{cases}$$

○ A: Starting from $\beta = 1$, Newton's method will take us to the minimum of f in one step

● B: Starting from $\beta = -1$, Newton's method will take us to the minimum of f in one step

● C: There exists a learning rate ϵ such that starting from $\beta = 1$, gradient descent will take us to the minimum of f in one step

● D: There exists a learning rate ϵ such that starting from $\beta = -1$, gradient descent will take us to the minimum of f in one step

A: The second derivative is 0 for $\beta \geq 0$, hence Newton's method will not be able to find the minimum starting at $\beta = 1$.

B: f is quadratic for $\beta \leq 0$ (the equality is not a typo), and the minimum of that quadratic is at $\beta = 0$, hence Newton's method will find the optimum starting from any $\beta < 0$ in exactly one step.

C: $\epsilon = 1$ works.

D: $\epsilon = 0.5$ works.

(g) [4 pts] In each of the following two figures, there are exactly three training points drawn from an unknown distribution, and the dashed line is a decision boundary.

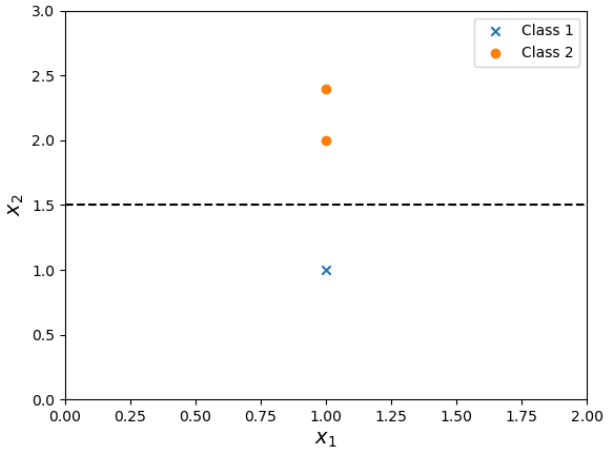


Figure 1

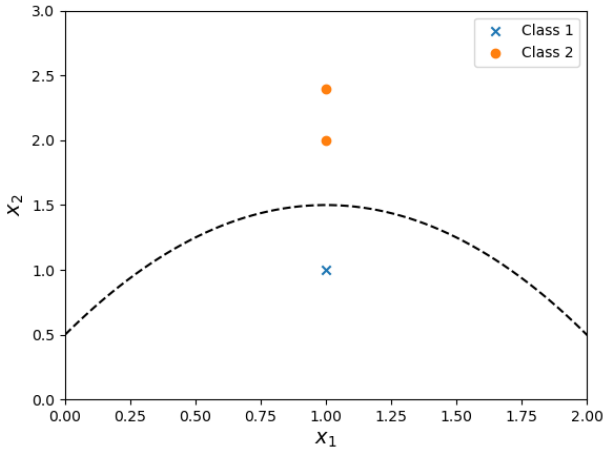


Figure 2

○ A: The Bayes optimal decision boundary always appears as drawn in Figure 1

● B: Both hard-margin and soft-margin (for some choice of C) SVMs could produce the decision boundary in Figure 1 (using only the features x_1 and x_2, plus the fictitious dimension where appropriate)

○ C: Both logistic regression and QDA could produce the decision boundary in Figure 2 (using only the features x_1 and x_2, plus the fictitious dimension where appropriate)

○ D: A hard-margin SVM augmented with the parabolic lifting map $\Phi(x) = [x_1 \quad x_2 \quad x_1^2 + x_2^2]^\top$ could produce the decision boundary in Figure 2

We cannot say anything about the Bayes optimal decision boundary from training points alone; we need to know the distributions that the training points are drawn from.

Logistic regression can only find linear decision boundaries when not given any features other than x_1 and x_2.

The parabolic lifting map cannot produce the decision boundary in Figure 2, because the boundary isn't a circle or a line. (Moreover, in this example, the parabolic lifting map would still produce a line.)

(h) [4 pts] Consider least-squares linear regression with a design matrix $X \in \mathbb{R}^{n \times d}$ and labels $y \in \mathbb{R}^n$. If the solution to the least-squares problem is unique, which of the following must be true?

- ● A: $\text{rank}(X) = d$
- ○ C: $n \leq d$
- ● B: $\text{rank}(X^\top) = d$
- ● D: $d \leq n$

The normal equations from the least-squares problem are

$$X^\top X w = X^\top y.$$

For the solution to be unique, we need to have that $X^\top X$ is invertible. This is equivalent to saying that $\text{rank}(X^\top X) = d$, and equivalently, $\text{rank}(X^\top) = d$. Since the rank of a matrix and the rank of its transpose must be that same, we must have that $\text{rank}(X) = d$ as well. Note that the rank of a matrix is always less than or equal to the number of columns, and also always less than or equal to the number of rows. In this case, regardless of the matrix X, we must have that $\text{rank}(X) \leq \min\{n, d\}$. Since we already determined that $\text{rank}(X) = d$, and $d \geq \min\{n, d\}$, we must therefore have that $d = \min\{n, d\}$ and thus $\text{rank}(X) = d = \min\{n, d\}$. Again, the rank of a matrix and that of its matrix is the same, so the same statement applies for $X^\top$.

(i) [4 pts] Suppose we are performing linear regression on a design matrix X and a label vector y. Recall that the conventional least-squares formulation finds the linear function $h(\cdot)$ that minimizes the empirical risk $R(h) = \frac{1}{n}\sum_{i=1}^{n} L(h(X_i), y_i)$, where $L(\zeta, \gamma) = (\zeta - \gamma)^2$ is the squared-error loss for a prediction $\zeta \in \mathbb{R}$ and a label $\gamma \in \mathbb{R}$. However, you are afraid that your training data may have outliers. Which of the following changes will help mitigate this issue if it exists?

● A: Change the loss function from $L(\zeta, \gamma) = (\zeta - \gamma)^2$ (squared error) to $L(\zeta, \gamma) = |\zeta - \gamma|$ (absolute error)

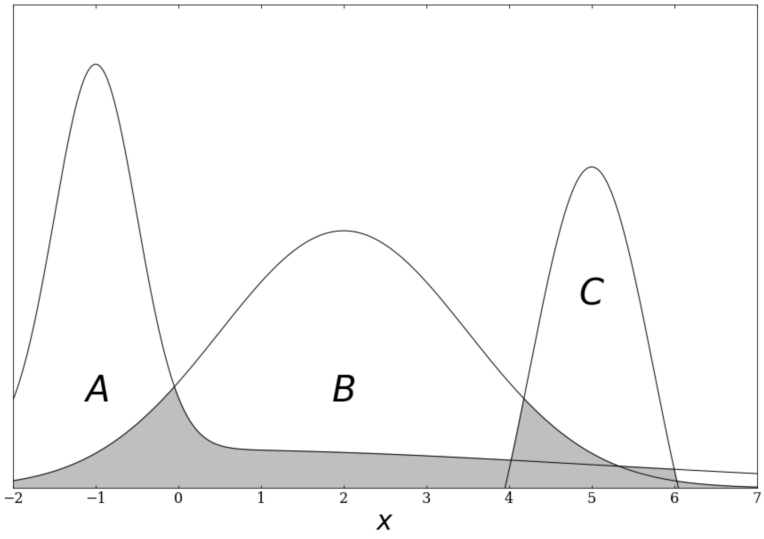
○ B: Change the loss function from $L(\zeta, \gamma) = (\zeta - \gamma)^2$ (squared error) to $L(\zeta, \gamma) = -\gamma \ln \zeta - (1 - \gamma) \ln (1 - \zeta)$ (logistic loss)

○ C: Change the cost function from $R(h) = \frac{1}{n}\sum_{i=1}^{n} L(h(X_i), \gamma_i)$ (mean loss) to $R(h) = \sum_{i=1}^{n} L(h(X_i), \gamma_i)$ (total loss)

○ D: Change the cost function from $R(h) = \frac{1}{n}\sum_{i=1}^{n} L(h(X_i), \gamma_i)$ (mean loss) to $R(h) = \max_{i \in [1,\ldots,n]} L(h(X_i), \gamma_i)$ (maximum loss)

Absolute error is linear wrt the error so it is less sensitive than squared error. Logistic loss is only valid for labels in $[0, 1]$ and is not robust to outliers anyways. Removing the $\frac{1}{n}$ in the objective does not change the optimization problem. Maximum loss is actually more sensitive to outliers than mean loss.

(j) [4 pts] The following chart depicts the class-conditional distributions $P(X|Y)$ for a classification problem with three classes, A, B, and C. Classes A and B are normally distributed over the domain $(-\infty, \infty)$; Class C is defined only over the finite domain depicted below. All three classes have prior probabilities π_A, π_B, π_C **strictly greater than zero**; the chart does **not** show the influence of these priors. We use the **0-1 loss** function.



○ A: The Bayes risk is the area of the shaded region in the chart (including the area not depicted off the sides of the chart, going to $x = \pm\infty$)

● B: Depending on the priors, it is possible that the Bayes rule $r^*(x)$ will classify all inputs as class B

○ C: Depending on the priors, it is possible that the Bayes rule $r^*(x)$ will classify all inputs as class C

○ D: Depending on the priors, it is possible that the Bayes risk is zero

A) This is wrong for several reasons. First, to get the Bayes risk, you would need to scale each curve by its prior. Second, some of the shaded area has to be counted twice; e.g., the area under A where B and C are both above A.

B and C) The only way for something like this to occur is if the prior for a class is significant enough that its joint distribution raises above the other classes' joint distribution for all x. Class C can't do that, as it does not span all of the x-space. Class B can, as it covers the whole number line and its variance is greater than Class A's.

D) Haha

Q1. [40 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [4 pts] Which of the following cost functions are smooth—i.e., having continuous gradients everywhere?

- ○ A: the perceptron risk function
- ● B: the sum (over sample points) of logistic losses
- ● C: least squares with ℓ_2 regularization
- ○ D: least squares with ℓ_1 regularization

(b) [4 pts] Which of the following changes would commonly cause an SVM's margin $1/\|w\|$ to shrink?

- ● A: Soft margin SVM: increasing the value of C
- ● B: Hard margin SVM: adding a sample point that violates the margin
- ○ C: Soft margin SVM: decreasing the value of C
- ○ D: Hard margin SVM: adding a new feature to each sample point

The greater the value of C is, the higher the penalty for violating the margin. The soft margin shrinks to compensate.

If you add a sample point that violates the margin, a hard margin always shrinks.

If you add a feature, the old solution can still be used (by setting the weight associated with the new feature to zero). Although the new feature might enable a new solution with a wider margin, the optimal solution can't be worse than the old solution.

(c) [4 pts] Recall the logistic function $s(\gamma)$ and its derivative $s'(\gamma) = \frac{\mathrm{d}}{\mathrm{d}\gamma} s(\gamma)$. Let γ^* be the value of γ that maximizes $s'(\gamma)$.

- ○ A: $\gamma^* = 0.25$
- ● B: $s(\gamma^*) = 0.5$
- ○ C: $s'(\gamma^*) = 0.5$
- ● D: $s'(\gamma^*) = 0.25$

A glance at the logistic curve and its symmetry makes it clear that the slope is maximized when $\gamma^* = 0$ and $s(\gamma^*) = 0.5$. Recall that $s' = s(1 - s)$; hence $s'(\gamma^*) = 0.25$.

(d) [4 pts] You are running logistic regression to classify two-dimensional sample points $X_i \in \mathbb{R}^2$ into two classes $y_i \in \{0, 1\}$ with the regression function $h(z) = s(w^\top z + \alpha)$, where s is the logistic function. Unfortunately, regular logistic regression isn't fitting the data very well. To remedy this, you try appending an extra feature, $\|X_i\|^2$, to the end of each sample point X_i. After you run logistic regression again with the new feature, the decision boundary in $\mathbb{R}^2$ could be

- ● A: a line.
- ● B: a circle.
- ○ C: an ellipse.
- ○ D: an S-shaped logistic curve.

The decision boundary is the set of points $\{z : s(w^\top z + \alpha) = 0.5\}$, which is the set of points $\{z : w^\top z + \alpha = 0\}$. If $z = [x_1 \quad x_2 \quad x_1^2 + x_2^2]$, then the decision boundary for points in x-space is $w_1 x_1 + w_2 x_2 + w_3(x_1^2 + x_2^2) + \alpha = 0$. This equation can express arbitrary circles and lines. It can't express ellipses because there is only one quadratic feature, and it causes x_1^2 and x_2^2 to always have the same coefficient. It definitely can't express any S-shaped curves.

(e) [4 pts] We are performing least-squares linear regression, with the use of a fictitious dimension (so the regression function isn't restricted to satisfy $h(0) = 0$). Which of the following will never increase the training error, as measured by the mean squared-error cost function?

- ● A: Adding polynomial features
- ○ B: Using backward stepwise selection to remove some features, thereby reducing validation error
- ○ C: Using Lasso to encourage sparse weights
- ● D: Centering the sample points

Adding new features never increases the training error. Centering the sample points never changes the training error (as we're using a fictitious dimension), as the change in coordinate system can be compensated for by a translation of the regression function (effected by a change in the bias term). Options B and C usually increase training error for the purpose of reducing validation error.

(f) [4 pts] Given a design matrix $X \in \mathbb{R}^{n \times d}$, labels $y \in \mathbb{R}^n$, and $\lambda > 0$, we find the weight vector w^* that minimizes $||Xw - y||^2 + \lambda||w||^2$. Suppose that $w^* \neq 0$.

- ● A: The variance of the method decreases if λ increases enough.
- ○ B: There may be multiple solutions for w^*.
- ● C: The bias of the method increases if λ increases enough.
- ○ D: $w^* = X^+ y$, where X^+ is the pseudoinverse of X.

A and C: as $\lambda \to \infty$, $w^* \to 0$ so the variance also approaches zero and the bias grows. When $\lambda > 0$, the objective is strongly convex and the minimizer is unique. D holds only when $\lambda = 0$, which we have said is not the case.

(g) [4 pts] **The following two questions use the following assumptions.** You want to train a dog identifier with Gaussian discriminant analysis. Your classifier takes an image vector as its input and outputs 1 if it thinks it is a dog, and 0 otherwise. You use the CIFAR10 dataset, modified so all the classes that are not "dog" have the label 0. Your training set has 5,000 dog images and 45,000 non-dog ("other") images. Which of the following statements seem likely to be correct?

○ A: LDA has an advantage over QDA because the two classes have different numbers of training examples.

○ B: QDA has an advantage over LDA because the two classes have different numbers of training examples.

○ C: LDA has an advantage over QDA because the two classes are expected to have very different covariance matrices.

● D: QDA has an advantage over LDA because the two classes are expected to have very different covariance matrices.

The "dog" class (label 1) only contains images of dogs, so compared to the "other" class (label 0), it should have less variance. The "other" class has images from many different classes, so it will have greater variance. QDA fits different covariance matrices to different classes, so it is more appropriate than LDA, which fits a single covariance matrix to both classes.

(h) [4 pts] **This question is a continuation of the previous question.** You train your classifier with LDA and the 0-1 loss. You observe that at test time, your classifier always predicts "other" and never predicts "dog." What is a likely reason for this and how can we solve it? (Check all that apply.)

● A: Reason: The prior for the "other" class is very large, so predicting "other" on every test point minimizes the (estimated) risk.

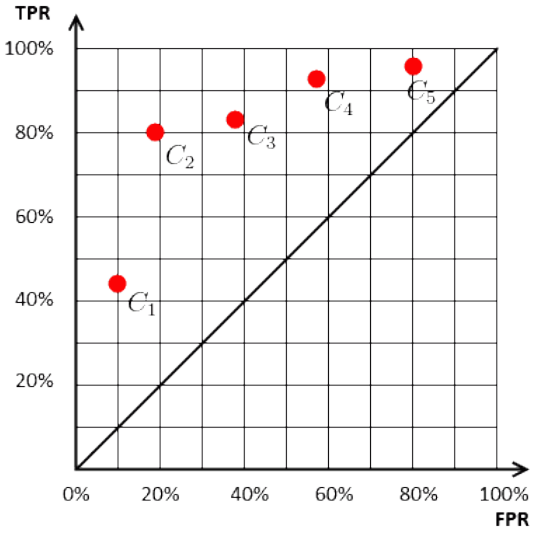
○ B: Reason: As LDA fits the same covariance matrix to both classes, the class with more examples will be predicted for all points in $\mathbb{R}^d$.

● C: Solve it by using a loss function that penalizes dogs misclassified as "other" more than "others" misclassified as dogs.

○ D: Solve it by learning an isotropic pooled covariance instead of an anisotropic one; that is, the covariance matrix computed by LDA has the form $\sigma^2 I$.

A: The 0-1 loss does not account for the imbalance in the number of examples per class, so it is minimized by predicting the class with the larger prior for a very large set of inputs if that prior is large enough (in this case, that set is large enough to contain the entire set of training images).
B: LDA can still predict the class with fewer examples when trained to minimize the 0-1 loss if the means of the two classes are different enough.
C: An asymmetric loss can counteract the larger prior on the "other" class.
D: lolno

(i) [4 pts] We do an ROC analysis of 5 binary classifiers C_1, C_2, C_3, C_4, C_5 trained on the training points X_{train} and labels y_{train}. We compute their true positive and false positive rates on the validation points X_{val} and labels y_{val} and plot them in the ROC space, illustrated below. In X_{val} and y_{val}, there are n_p points in class "positive" and n_n points in class "negative." We use a 0-1 loss.



ROC analysis of five classifiers. FPR = false positive rate; TPR = true positive rate.

● A: If $n_p = n_n$, C_2 is the classifier with the highest validation accuracy.

● B: If $n_p = n_n$, all five classifiers have higher validation accuracy than any random classifier.

● C: There exists some n_p and n_n such that C_1 is the classifier with the highest validation accuracy.

○ D: There exists some n_p and n_n such that C_3 is the classifier with the highest validation accuracy.

If $n_p = n_n$, the validation accuracy of each classifier is 0.5(True Positive Rate) +0.5(True Negative Rate) and C_2 is the best classifier, with an accuracy of about 80%. All five classifiers lie above the diagonal line, so they all have validation accuracy better than 50%, whereas any random classifier (biased or not) will have a validation accuracy of exactly 50% when $n_p = n_n$. C_1 is the best classifier when nearly all the validation data is negative, but C_3 is never the best classifier for any n_p/n_n, as it lies below the ROC convex hull.

(j) [4 pts] Tell us about feature subset selection.

○ A: Ridge regression is more effective for feature subset selection than Lasso.

○ B: If the best model uses only features 2 and 4 (i.e., the second and fourth columns of the design matrix), forward stepwise selection is guaranteed to find that model.

○ C: Stepwise subset selection uses the accuracy on the training data to decide which features to include.

● D: Backward stepwise selection could train a model with only features 1 and 3. It could train a model with only features 2 and 4. But it will never train both models.

A: Lasso can be used for subset selection because it encourages weights of zero. Ridge regression almost never produces weights of exactly zero.

C: Subset selection uses validation accuracy, not training accuracy.

B: Stepwise subset selection can miss the best subset. If the best 1-feature model uses only feature 3, then forward stepwise selection won't investigate the model that uses features 2 and 4.

D: It will never train both because it only tries training 2-feature models when it has already reduced the number of features to three, and those three features cannot contain all of features 1, 2, 3, and 4.

Q1. [48 pts] Multiple Answer

Fill in the bubbles for **ALL correct choices**: there may be more than one correct choice, but there is always at least one correct choice. **NO partial credit**: the set of all correct answers must be checked.

(a) [4 pts] Select the true statements about Bayes decision theory.

○ A: The risk for a decision rule is the average loss over the training points that are in class C.

○ B: The Bayes decision boundary between two classes, if you're using the 0-1 loss, is the set of points x where $P(X = x|Y = 0) = P(X = x|Y = 1)$.

● C: If the Bayes risk is nonzero in a two-class classification problem, then the distributions for each class (i.e., $P(X|Y = C)$ and $P(X|Y \neq C)$) must overlap.

● D: There exists a loss function for which the Bayes decision rule might select the class with lower posterior probability.

A: False, it is the expected loss over all values of x and y, and it has nothing to do with training points.

B: False; it's the set of points x where $P(Y = 0|X = x) = P(Y = 1|X = x)$.

C: True, when the class distributions are non-overlapping, each x will uniquely lie in one classes distribution, which means there exists a (optimal) classifier which perfectly classifies all points, and has Bayes risk of zero. Similarly, if Bayes risk is nonzero, there must be uncertainty for an x, and thus the class distributions must overlap.

D: True, for a nonsymmetric loss function.

(b) [4 pts] Select the true statements about least-squares linear regression.

○ A: The problem of minimizing $\|Xw - y\|_1$ often yields a "sparse" solution, where some of the components of w are exactly zero.

● B: There is always at least one solution to the normal equations.

○ C: There are problems for which the normal equations have exactly two distinct solutions.

○ D: When the normal equations have multiple solutions, all the solutions have the same loss on test points.

A is false. Minimizing $\|w\|_1$ encourages weights to be exactly zero, but minimizing $\|Xw - y\|_1$ does not.
B is true. (Proved in discussion section.)
C is false. (See discussion section.)
D is false. The different solutions give different values when you move off the affine subspace that all the training points lie on. Those different values will give different losses on test points.

(c) [4 pts] Select the true statements about ROC curves.

○ A: The horizontal axis represents posterior probability thresholds and the vertical axis represents test set accuracy.

● B: The ROC curve is a better guide for choosing a threshold (separating negative from positive classifications) on real-world data than the threshold suggested by decision theory.

○ C: A ROC curve closer to the diagonal line $y = x$ implies that your classifier's risk is closer to Bayes optimal.

● D: There are (at least) two points on a ROC curve that are not affected by changes in the model. (Note: we are not counting the specific choice of threshold between positive and negative as part of the model).

A is false because both axes of the ROC curve show classification rates. The possible classifier thresholds are implicitly-not explicitly- shown on the ROC curve. B is true (see lecture note 11) C is nearly the opposite of the truth, because the line $y = x$ represents a random classifier. D is true (the points (0,0) and (1,1) are always present since they arise when we only classify as one class).

(d) [4 pts] Ridge regression is

○ A: a way to perform feature selection, as ridge regression encourages weights to be exactly zero.

● B: a method in which bias tends to increase, and variance tends to decrase, as we increase the regularization parameter λ.

● C: motivated by imposing a Gaussian prior probability on the weight vector.

● D: a method whose cost function has a unique minimum (assuming $\lambda > 0$).

A: Lasso does that, but ridge regression rarely produces weights of exactly zero.

B: True. As $\lambda \to \infty$, the weights approach zero, and so does the variance, whereas the bias increases.

C: True. Another way to derive ridge regression is via *maximum a posteriori* with a Gaussian prior on the weights.

D: True. Ridge regression's regularization term makes the objective function strictly convex.

(e) [4 pts] Select the statements that are true for **every** real symmetric matrix $X \in \mathbb{R}^{n \times n}$.

● A: X can be factored as $X = UDU^\top$, where U is a orthogonal matrix and D is a diagonal matrix.

○ B: X can be factored as $X = UU^\top$, where U is a orthogonal matrix.

○ C: $\lambda_{\max}(X) \geq 0$, where $\lambda_{\max}(X)$ denotes the greatest eigenvalue of X.

● D: $a^\top X a \leq \lambda_{\max}(X) \|a\|_2^2$ for all $a \in \mathbb{R}^n$.

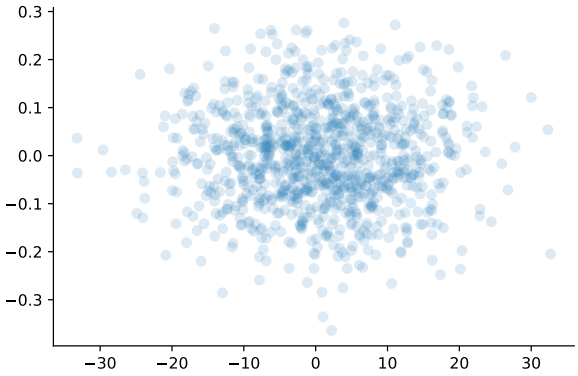
A: True, this is a direct statement of spectral theorem.

B: Not true, as this entails X being PSD.

C: Not true, X can be $-I_n$.

D: True. Proof can be found in HW2.

(f) [4 pts] Below are 1,000 sample points drawn from a two-dimensional multivariate normal distribution. Which of the following matrices could (without extreme improbability) be the covariance matrix of the distribution? (Pay attention to the numbers on the axes!)



● A: $\Sigma = \begin{bmatrix} 100 & 0 \\ 0 & 0.01 \end{bmatrix}$

○ C: $\Sigma = \begin{bmatrix} 10 & 0 \\ 0 & 0.1 \end{bmatrix}$

○ B: $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

○ D: $\Sigma = \begin{bmatrix} -10 & 0 \\ 0 & -0.1 \end{bmatrix}$

A is valid. B would be spherical. C would not have the right coordinates. Remember, the widths of the Gaussian are the standard deviations in each eigevector direction; and the standard deviations are the square roots of the variances, which are the eigenvalues of Σ. D cannot be a covariance matrix.

(g) [4 pts] You are training a soft-margin SVM on a binary classification problem. You find that your model's training accuracy is very high, while your validation accuracy is very low. Which of the following are likely to improve your model's performance on the validation data?

● A: Training your model on more data.

○ C: Increasing the hyperparameter C.

○ B: Adding a quadratic feature to each sample point.

● D: Decreasing the hyperparameter C.

A is true as training on more data reduces variance and decreases overfitting in general. B is false since polynomial features increase the variance and the risk of overfitting. C is false because increasing C enforces a harder margin constraint, leading to more overfitting. D is true because decreasing C allows for more slack, decreasing the risk of overfitting.

(h) [4 pts] Select the true statements about Gaussian Discriminant Analysis.

○ A: If a class-conditional covariance matrix is anisotropic (the eigenvalues are not equal), the decision boundary is guaranteed to be nonlinear.

● C: QDA is more prone to overfitting than LDA.

● B: The QDA posterior probability is a logistic function composed with (applied to) a quadratic function of the feature space.

○ D: The Bayes decision boundary arising from two normally distributed classes can split the feature space into *at most* two regions.

A is False; if all classes have the same anisotropic covariance matrix, the decision boundary is linear. B is true, and can be shown by Bayes' rule. C is true since QDA has a higher number of parameters than LDA. D is false, since the decision boundary can split the feature space into three regions if the class covariances are not equal.

(i) [4 pts] Select the true statements about finding a minimum of a cost function $f(x)$.

○ A: Newton's method always converges to a globally minimum solution for any twice-differentiable function f.

● B: For the cost function $f(x) = \delta\|x - b\|^2 + \gamma$ with $\delta > 0$, Newton's method always converges to a globally minimum solution.

○ C: If f is convex, is differentiable, and has exactly one local minimum, then (batch) gradient descent always converges to that minimum for any choice of learning rate.

● D: It is not possible to execute an iteration of Newton's method on the perceptron risk function.

A is false: Newton's method doesn't always converge, and when it does, the critical point it finds isn't always a minimum.
B is true as the globally minimum solution is b and $\nabla f(x) = 2\delta(x-b)$ and $\nabla^2 f(x) = 2\delta I$ and $x_1 = x_0 - \frac{2\delta(x_0-b)}{2\delta} = x_0 - (x_0 - b) = b$
C is false as gradient descent will not converge if the learning rate is too high.
D is true because the perceptron risk function has a zero Hessian matrix everywhere, except at points where the Hessian isn't even defined.

(j) [4 pts] In the following statements, the word "bias" is referring to the bias-variance decomposition. Select the true ones.

○ A: A model trained with n training points is likely to have lower variance than a model trained with $2n$ training points.

● B: If my model is underfitting, it is more likely to have high bias than high variance.

○ C: Increasing the number of parameters (weights) in a model usually improves the test set accuracy.

● D: Adding ℓ_2-regularization usually reduces variance in linear regression.

A is false since increasing the number of samples generally reduces the variance.
B is generally true, since you typically underfit if you have insufficient model capacity to even perform well on the training set.
C is not always true, as increasing the number of parameters can lead to overfitting.
D is true, regularization effectively reduces model complexity.

(k) [4 pts] Which of the following statements are true regarding Lasso regression?

● A: Lasso's optimization problem can be stated as a quadratic program.

● B: The cost function minimized by Lasso has points where its gradient is not well-defined, and the solution (minimum) is often at such a point.

● C: Lasso often produces sparser results (more zero weights) than ridge regression.

● D: A version of Lasso using a penalty term of $\lambda\|w\|_{\ell_{0.5}}$ (that is, the $\ell_{0.5}$-norm) will be more inclined to produce sparse solutions than Lasso.

A is not true in general. Lasso *tends* to produce sparser solutions, but there are cases where it may produce equivalently sparse solutions as other forms of regression.
B is false because Lasso's objective function is nondifferentiable.
C is true. This is due to the square-shaped isocontours introduced by the ℓ_1 norm being more likely to intersect other isocontours on axes than the circular isocontours introduced by the ℓ_2 norm.
D is true — intuitively, the smaller the p we select for our p-norm, the pointier the penalty's isocontours become, making us more likely to choose a sparse weight vector.

(l) [4 pts] Let X be an $n \times d$ design matrix where $n = 10$ and $d = 12$, representing information about various loan borrowers. Let $y \in \mathbb{R}^n$ be a vector of labels such that y_i represents the time (in days) between when borrower i took a loan and when it was fully repaid. We would like to train a regression model on this data. Which of the following methods would be reasonable choices for this task?

○ A: Least squares linear regression with the solution $w^* = (X^\top X)^{-1} X^\top y$

○ B: Logistic regression

● C: Least squares linear regression using the Moore–Penrose pseudoinverse, $w^* = X^\dagger y$

● D: Ridge regression

A: This does not work because $d > n$, meaning $X^\top X$ will be rank-deficient and thus not invertible.

B: Logistic regression is limited to predicting values between 0 and 1, which is not appropriate for our use case (predicting time in days).

C: Ridge regression avoids the problem with (A) because the matrix $X^\top X + \lambda I$ will always be invertible.

D: As we saw in discussion, the Moore-Penrose pseudoinverse allows us to find the unique least norm least squares solution even if $X^\top X$ is non-invertible.